INCH-POUND

MIL-DTL-32262B (MR)
21 April 2025
SUPERSEDING
MIL-DTL-32262A (MR)
16 March 2015

DETAIL SPECIFICATION

ARMOR PLATE, ALUMINUM ALLOY 6055 WELDABLE & ALLOY 6061, UNWELDABLE APPLIQUE

This specification is approved for use by the Department of the Army and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers two wrought aluminum alloy armor plate for both welded and non-welded applications in nominal thicknesses from 0.250 to 3.000 inch, inclusive (see 6.2). The weldability of wrought aluminum alloy AA6055 armor has been determined for class II armor for this thickness only; 1.500". The weldability of AA6061 wrought aluminum alloy armor has not been determined. Therefore, this alloy should only be used as appliqué armor.

1.2 Weldability. Class II material covered by this specification has been demonstrated to be weldable to itself and other weldable alloys (see 6.3).

1.3 Classification. The wrought aluminum armor should be of the following classes as specified (see 6.2).

1.3.1 Class I. Class I is wrought aluminum armor that conforms to the Aluminum Association designation for the 6061 aluminum alloy. The applicable gauge range for class I is 0.250 - 3.000 inches.

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory (ARL), Army Research Directorate, Specifications & Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9535

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1.3.2 Class II. Class II is wrought aluminum armor that conforms to the Aluminum Association designation for the 6055 aluminum alloy. The applicable gauge range for class II is 0.500 - 3.000 inches.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662	-	V50 Ballistic Test for Armor
MIL-STD-3057	-	Arc Welding of Armor Grade Aluminum

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

TEST OPERATING PROCEDURES

TOP 2-2-711	-	Proofing Projectile Impact Testing of Armor Weldments

(Copies of these documents are available online at https://quicksearch.dla.mil/).

2.3 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

THE ALUMINUM ASSOCIATION, INC.

Teal Sheets	-	International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys

(Copies of these documents are available online at https://www.aluminum.org/).

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THE AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI H35.2	-	American National Standard Dimensional Tolerances for Aluminum Mill Products

(Copies of these documents are available online at https://ansi.org/).

ASTM INTERNATIONAL

ASTM B557	-	Standard Test Methods for Tension Testing Wrought and Cast Aluminum and Magnesium-Alloy Products
ASTM E716	-	Standard Practices for Sampling and Sample Preparation of Aluminum and Aluminum Alloys for Determination of Chemical Composition by Spark Atomic Emission Spectrometry
ASTM E1251	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Spark Atomic Emission Spectrometry
ASTM E3061	-	Standard Test Method for Analysis of Aluminum and Aluminum Alloys by Inductively Coupled Plasma Atomic Emission Spectrometry (Performance Based Method)
ASTM G38	-	Standard Practice for Making and Using C-Ring Stress-Corrosion Test Specimens
ASTM G47	-	Standard Test Method for Determining Susceptibility to Stress-Corrosion Cracking of 2XXX and 7XXX Aluminum Alloy Products

(Copies of these documents are available online at https://www.astm.org/.)

SAE INTERNATIONAL

AMS 2750	-	Pyrometry
AMS 2772	-	Heat Treatment of Aluminum Alloy Raw Materials

(Copies of these documents are available online at https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), first article testing shall be required and all test samples required by this specification shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3. First article testing shall be completed before production material is submitted for acceptance testing. The approval of the first article samples authorizes commencement of production but does not relieve the supplier of the responsibility to comply with all the applicable provisions of this specification. The first article samples and acceptance test plates shall be manufactured by the process proposed for use on production items.

3.2 Chemical composition. The chemical compositions for classes I and II alloys defined by the Aluminum Association Teal Sheets, "International Alloy Designations and Chemical Composition Limits for Wrought Aluminum and Wrought Aluminum Alloys" are listed in table I and shall be within those limits. A certification of conformance of the chemical composition of the alloy shall be furnished with the ballistic test plates.

TABLE I. Chemical composition, percent. 1/

ELEMENTS	SYMBOL	Class I 6061 ALLOY 2/	Class II 6055 ALLOY 2/
Silicon	Si	0.40 - 0.80	0.60 - 1.2
Iron	Fe	0.70	0.30
Copper	Cu	0.15 - 0.40	0.50 - 1.0
Manganese	Mn	0.15	0.10
Magnesium	Mg	0.8 - 1.2	0.70 - 1.1
Chromium	Cr	0.04 - 0.35	0.20 - 0.30
Zinc	Zn	0.25	0.55 - 0.90
Titanium	Ti	0.15	0.10
Other, max. Each	---	0.05	0.05
Other, max. Total 3/	---	0.15	0.15
Aluminum	Al	Remainder	Remainder

1/ Except for "Aluminum" and "others", analysis normally is made for elements for which specific limits are shown.
2/ Where single units are shown, these indicate the maximum amounts permitted.
3/ The sum of those "others" metallic elements 0.010 percent or more each, expressed to the second decimal before determining the sum.

3.3 Mechanical properties. Unless otherwise specified in the contract or order (see 6.2), the mechanical properties of the test specimen taken in the longitudinal (L) direction shall meet the minimum mechanical properties listed in table II. If mechanical property requirements differ from those contained in table II, or if any other properties are required, the ballistic requirements shall be negotiated between the procuring activity and the supplier.

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TABLE II. Minimum mechanical properties. [1/] [2/]

Thickness, inches	Tensile Strength, Ksi		Yield Strength, 0.2% Offset, Ksi		Elongation percent	
	Class I	Class II	Class I	Class II	Class I	Class II
0.250 to 0.499, incl.	38.0	N/A	35.0	N/A	10	N/A
0.500 to 2.000, incl.	38.0	52	37.0	50	10	6
2.001 to 3.000, incl.	38.0	52	35.0	50	10	5

[1/] The test specimen gage length shall be 1.400 inch for plates having a nominal thickness of 0.500 inch; all other thicknesses shall have a gage length of 2.000 inches.
[2/] Values are taken in the Longitudinal Direction (see 3.3).

3.4 Weldability. When the vehicle requires weldments, the fabricators (OEMs and kit manufacturers, etc.) are required to demonstrate weldability in accordance with MIL-STD-3057, "Arc Welding of Armor Grade Aluminum" for all armor to armor and armor to non-armor applications. This may also require Proofing Projectile Impact Testing (PPIT) for armor to armor weldments in accordance with procedures in TOP 2-2-711, "Proofing Projectile Impact Testing of Armor Weldments" (revision A and above) depending on the weld classification in accordance with MIL-STD-3057. PPIT procedures and associated striking velocities for established welded class II plate thicknesses are listed in TOP 2-2-711. For any newly introduced armor material, configuration, or unproven thickness being used for armor applications, the aluminum fabricators shall qualify a weld procedure in accordance with MIL-STD-3057, and this requires PPIT testing following TOP 2-2-711. Since ordered thicknesses of the alloys referenced by this specification may not be listed in TOP 2-2-711, it is strongly suggested that the contract or drawing for the vehicle specify weldability requirements as outlined in paragraph 6.3.

3.5 Ballistic limit. The protection ballistic limit, BL(P), shall be as specified in Appendix A. No ballistic limit for class I material shall be specified for plate thickness of 0.250 to 0.499 inch, inclusive. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.6 Stress corrosion resistance. Stress Corrosion Cracking resistance is an optional requirement under this specification. When stress corrosion testing is requested in the contract or purchase order (see 6.2) and unless otherwise specified in the contract or purchase order (see 6.2), plate 0.750 inch and over in thickness shall be resistant to stress corrosion cracking, with a minimum of 5 of the 9 specimens showing no evidence of cracking at the end of 96 hours when tested as specified in 4.7.5.

3.7 Thermal processing. Heat treatment shall conform to the requirements of SAE AMS 2750 and SAE AMS 2772 and shall be such as to enable the material to meet the requirements of these specifications. Any thermal processing exceeding 200°F (93°C) excluding fabrication by welding in accordance with MIL-STD-3057 shall necessitate recertification for conformance to 3.3 and 3.4.

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3.8 Dimensions. Dimensions for plates delivered for fabrication shall have an overall dimension tolerance of +0.500/-0.000 for width and length unless otherwise specified in the contract or order (see 6.2).

3.8.1 Tolerances. Unless otherwise specified in the contract or order (see 6.2), the plates shall not vary from the specified ordered dimensions by an amount greater than that specified by ANSI H35.2, except for thickness.

3.8.2 Thickness. Thickness tolerance for production armor and ballistic test plates shall be as specified in table III.

TABLE III. Thickness tolerances.

Ordered Thickness (Inches)		SPECIFIED WIDTH (Inches)							
	OVER	0.00	39.37	59.06	78.74	98.43	118.11	137.80	157.48
	THRU	39.37	59.06	78.74	98.43	118.11	137.80	157.48	177.17
OVER	THRU	TOLERANCES - INCHES (PLUS and MINUS)							
0.500	1.000	0.031	0.031	0.037	0.043	0.051	0.060	0.070	0.085
1.000	1.575	0.039	0.039	0.047	0.055	0.065	0.075	0.090	0.105
1.575	2.362	0.055	0.055	0.060	0.070	0.085	0.100	0.115	---
2.362	3.000	0.075	0.075	0.085	0.100	0.105	0.125	---	---
3.000	4.000	0.090	0.090	0.105	0.110	0.125	---	---	---

3.9 Marking for identification. Unless otherwise specified in the contract or purchase order (see 6.2) each plate shall be marked on one plate edge with the manufacturer's name or CAGE code, the basic number of this specification, the plate thickness in inches, the alloy designation and the lot number or code relating to the lot number (see 4.2). The height of the characters shall be 3/8 of an inch or greater. Impression stamping shall not be used unless permitted by the procuring activity (see 6.2). Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall not contain more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

3.9.1 Ballistic test plates. In addition to the markings in 3.8, each ballistic test plate shall be marked with the letters PRE for First Article test plates and ACC for Acceptance test plates. This marking shall be impression stamped on the edge in letters 3/8 inch high or greater. Ballistic retest plates shall be marked "R1" and "R2" respectively (see A.5.2).

3.10 Ballistic test plate information. For each lot of aluminum alloy armor, a properly completed Aluminum Armor Test Data Form (See figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.11 Workmanship. Plate produced under this specification shall be uniform in quality and clean, smooth and sufficiently free from buckles, blisters, hard spots, damaged ends, laminations and other defects which may affect its use. Minor scratches, gouges, dents and similar damage (see

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3.11.2) are acceptable if the thickness is not reduced below the minimum material tolerance thickness.

3.11.1 Surface and Edge Condition. The surface and edge condition of the plate shall be free of surface cracks, edge cracks, and edge laminations as defined in the contract or purchase order (see 6.2).

3.11.2 Imperfections. The depth of surface scratches, nicks, pits, or gouges shall not reduce the plate thickness below the allowable minimum thickness. Isolated individual pits over 0.015 inches deep but not over 0.030 inches deep and not within 6 inches of each other and do not violate the minimum allowable thickness, are acceptable.

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REQUEST FOR BALLISTIC TEST OF ALUMINUM ALLOY ARMOR													
SPECIFICATION: MIL-DTL-32262				**REVISION: B**				**AMENDMENT:**					
FIRING RECORD:			**DATE:**			**TEST CONTRACT NO:**							
	PLATE MANUFACTURER / PRODUCER					**PRIME CONTRACTOR**							
Name:													
Address:													
POC:													
Phone No:													
Email:													
TEST ITEM IDENTIFICATION:						Ordered Thickness:							
Lot No. _____ **Plate No.** _____				**Alloy:** _____ **Temper:** _____				**Class:** _____ **Type:** _____					
PURPOSE: __ Acceptance ___ First Article ___ Development ___ Reduced Testing ___ Audit Testing													
SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample _____)													
CHEMISTRY ANALYSIS:	**Si**	**Fe**	**Cu**	**Mn**	**Mg**	**Cr**	**Zn**	**Ti**	**Other - Each**			**Other - Total**	**Al**
													Rem.
MECHANICAL PROPERTIES:													
UTS (ksi):				0.2% YS (ksi):					Elongation (%):				
WELDABILITY :	Thickness			Required Velocity			Actual Velocity			Pass/Fail			
BALLISTIC TEST RESULTS: **SEPARATELY HEAT TREATED TEST PLATE : YES ____ NO ____**													
Test: **Projectile**:	**Obl. (deg)**		**Actual Thickness (in)**			**Required V_{50} (fps)**			**Actual V_{50} (fps)**			**Pass/ Fail**	
	NOTES:												
SPALL:	Spall Size:					Remarks:							
DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT & SUPPLIER OFFICIALS													
LOT: [____ met] [____ failed to meet] the ballistic requirements of specification MIL-DTL-32505													
Government Representative:			Date:			Supplier Representative:						Date:	

Figure 1. Aluminum Armor Test Data Form.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Lot. A lot shall consist of all plate(s) of the same alloy and ordered thickness which has been processed together by the same mill practice. Unless otherwise specified in the contract or purchase order (see 6.2), the weight of the finished plate(s) in the lot shall not exceed 50,000 pounds and shall be submitted for inspection as a unit. All lots shall be fully traceable back to the ingot source by the supplier.

4.3 First article. When specified in the contract or purchase order, (see 6.2) and before production has commenced, samples of specified material shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.3.1. The approval of the first article samples authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification The first article samples shall be produced using the mill production practice proposed for use on production.

4.3.1 First article inspection. First article inspection except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the conformance inspection or production acceptance inspection shown in 4.4.

4.4 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examinations of 4.6 and the tests of 4.7.

4.5 Sampling.

4.5.1 First article inspection.

4.5.1.1 Chemical composition. One (1) sample for chemical analysis shall be removed from each plate selected for ballistic testing and shall meet the requirements of 3.2 when tested as specified in 4.7.1.

4.5.1.2 Mechanical properties. One tension test specimen shall be removed from each plate that has been selected for ballistic testing and shall meet the requirements when tested as specified in 4.7.2.

4.5.1.3 Ballistic tests. Two plates, 12 inches by 36 inches of each thickness to be supplied on the contract, shall be submitted for ballistic testing in accordance with Appendix A. The orientation of these plates with respect to the rolling direction shall be at the option of the producer (see 6.2).

4.5.1.4 Stress corrosion tests. When required (see 6.2), nine (9) stress corrosion test specimens shall be removed from the same plate that has been selected for ballistic testing. The location and type of specimen shall be as specified in 4.7.5.

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4.5.2 Conformance inspection.

4.5.2.1 Chemical composition. The sample shall meet the chemical composition requirements of 3.2 when tested as specified in 4.7.1.

4.5.2.1.1 Ingot analysis. At least one sample shall be taken from the molten metal representing the ingots poured as a unit from the same source molten metal. Complete ingot analysis records shall be available to the Government at the producer's facility.

4.5.2.1.2 Product analysis. When sampling has not been made in accordance with 4.5.2.1.1, one sample shall be randomly taken for each 4,000 pounds or less in a lot. Complete product analysis records shall be available to the Government at the contractor's facility.

4.5.2.2 Mechanical properties. Samples for tension tests shall be selected from each lot in accordance with table IV. Each sample shall be randomly selected from a different plate in the lot, and only one tension test specimen shall be made from each sample. The sample shall meet the requirements when tested as specified in 4.7.2.

TABLE IV. Number of tension tests.

Lot size, pounds	Minimum number of samples 1/
To 8,000, incl.	2
8,001 to 12,000, incl.	3
12,001 to 20,000, incl.	4
20,001 up	5

1/ If a lot consists of only one plate, one sample shall be required.

4.5.2.3 Ballistic testing. One plate, 12 inches by 36 inches, shall be randomly selected from each lot for ballistic testing. The orientation of the plate with respect to the rolling direction shall be at the option of the producer (see 6.2). The sample shall meet the requirements when tested as specified in 4.7.4.

4.5.2.4 Stress corrosion tests. When required (see 6.2), nine test specimens shall be prepared from one plate randomly selected from each lot for stress corrosion testing. The plate shall meet the requirements when tested as specified in 4.7.5.

4.6 Examination.

4.6.1 Visual. Each plate shall be examined for compliance with the identification marking (see 3.9) and workmanship (see 3.11) requirements.

4.6.2 Dimensions. Plates within a lot shall be measured to determine compliance with requirements of paragraph 3.8 in accordance with the sampling procedures approved by the procuring activity and as specified in the contract or purchase order (see 6.2). At a minimum, one plate per lot shall be randomly selected for dimensional inspection.

4.7 Test specimens.

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4.7.1 Chemical composition. Samples for chemical analysis shall be prepared and tested in accordance with one or more ASTM methods of E716, E1251, and E3061. In case of dispute, analysis by method E3061 shall be the basis for acceptance or rejection.

4.7.2 Mechanical properties. Tension test specimens shall be prepared and tested in accordance with ASTM B557. Specimens shall be taken in the longitudinal (L) direction. For material less than 0.500 inch in thickness, a standard rectangular tension test specimen shall be used. For plate in nominal thickness 0.500 to 1.500 inches, inclusive, tension test specimens shall be taken with the axis midway between the two plate surfaces. For plate in nominal thickness greater than 1.500 inches, the axis of the tension test specimen shall be three-fourths of the distance from one surface to the other.

4.7.3 Weldability. When required (see 6.2), the thickness of the weldments specified in the contract or drawing shall be used to determine the proofing projectile and striking velocity in accordance with MIL-STD-3057 and TOP 2-2-711.

4.7.4 Ballistic testing. The ordered thickness specified in the contract or purchase order shall be used to determine the test projectile in accordance with table V. Ballistic testing shall be in accordance with Appendix A. Test plate thickness, as measured by the ballistic testing agency, shall be used in conjunction with table V and Appendix A to determine the required V_{50} protection ballistic limit for that plate. Thickness shall be determined as the average of at least four thickness measurements read on a deep throat micrometer or by means of an ultrasonic device to the nearest 0.001 of an inch and rounded off to the nearest 0.005 of an inch. Measurements shall be made on the intended impact area. In those cases where the BL(P) is within ±10 fps of the minimum required value for the measured average thickness (to the nearest 0.005-inch), an interpolation of the appropriate ballistic limit table shall be performed. The average plate thickness, computed to the nearest 0.001-inch, shall be used to determine the minimum required BL(P) for that plate.

TABLE V. Acceptance ballistic test plates.

Ordered Thickness, Inches	Projectile	Angle of Obliquity in Degrees	Class I (6061)	Class II (6055)
0.500 - 0.749	Cal. .30 AP M2	30	Table A-I	Table A-I
0.750 - 0.950 0.750 - 1.000	Cal. .50 FSP	0	Table A-II	Table A-II
0.951 - 1.500 1/ 1.001 - 1.700 2/	20-mm FSP	0	Table A-III	Table A-III
0.951 - 1.500 1/ 1.001 - 1.700 2/ 1.701 - 2.000	Cal. .30 AP M2	0	Table A-IV	Table A-IV
1.501 - 3.000 2.001 - 3.000	Cal. .50 AP M2	0	Table A-V	Table A-V

1/ Two (2) types of projectiles for the same thickness range.
2/ Two (2) types of projectiles for the same thickness range.

4.7.4.1 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to the Commander, USA ATC, ATTN: TEDT-AT-

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SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance.

4.7.4.2 Incomplete penetrations. When a complete penetration cannot be obtained, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

4.7.5 Stress corrosion specimens. When stress corrosion testing is required (see 6.2), short transverse stress corrosion test specimens shall be prepared in accordance with ASTM G38 and tested in accordance with ASTM G47. Nine specimens shall be tested per sample per lot. The specimens shall be stressed to 25 ksi. A maximum delay of 3 hours between stressing and initiation of stress corrosion test is permitted.

4.8 Rejection and retest. Unless otherwise specified in the contract or order (see 6.2) and except as specified in 4.7.2 and 4.7.3, rejection and retest shall be conducted in accordance with 4.8.1, 4.8.1.1, 4.8.2, and 4.8.3.

4.8.1 Rejection of first article plates. When one or more first article test specimens fail to meet the requirements of 4.3, the product lot and process, represented by the test plates or specimens shall be subject to rejection except as otherwise provided in a sampling plan approved by the procuring activity and in requirements of 4.8.1.1.

4.8.1.1 Retest of first article samples. Resubmission and retest of first article samples shall not be made until the manufacturer has made necessary corrections in the processing of the material to the satisfaction of the procuring activity. If one of the retest specimens fails, the lot shall be permanently rejected with no further testing permitted.

4.8.2 Ballistic retest. Rejection and retest of ballistic test plates shall be in accordance with A.5.2.

4.8.3 Stress corrosion retest. Immediately upon notification of the failure of the stress corrosion specimens, the manufacturer may at their expense test thirteen additional specimens for retesting. A minimum of seven specimens out of the thirteen additional specimens shall show no cracks upon examination at the end of 96 hours when tested as specified in 4.7.5. If the retest also fails, the manufacturer may elect to resubmit the lot after retreatment of the entire lot. After retreatment, the lot must pass the requirements outlined in 4.7.2, 4.7.4 and 4.7.5.

4.9 Reduced testing. At the discretion of the procuring activity and as specified in the contract or purchase order (see 6.2), the amount of testing may be reduced provided the results on consecutive lots indicate that a uniform product meeting the testing requirements is being produced and providing the manufacturer agrees to maintain the same manufacturing procedures. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat vehicles to protect the occupants against small arms fire, fragments, and shrapnel. Application for the AA6061 alloy is the M1114 Up-Armored HMMWV vehicle. Program Executive Offices (PEO) for the Ground Combat Systems will also use AA6061 alloys for their applications. Applications for the AA6055 alloy are as a replacement for AA6061 where increased AP ballistic performance or weldability is required.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Specify classification of alloy, and ordered thickness (see 1.1 and 1.3).
(c) When first article is required (see 3.1 and 4.3).
(d) Special mechanical properties (to include thermal stability mechanical properties) and ballistic requirements if different (see 3.3).
(e) When stress corrosion testing is required and if the test conditions are different (see 3.6, 4.5.1.4, 4.5.2.4, and 4.7.5).
(f) Dimension and tolerance requirements, if different (see 3.8 and 3.8.1).
(g) If markings are different and when impression stamping is permissible (see 3.9).
(h) Define surface cracks, edge cracks and edge laminations for the specific application (see 3.11.1).
(i) If the weight of finished plate can exceed 50,000 pounds (see 4.2).
(j) The orientation of the ballistic plate with respect to rolling is different (see 4.5.1.3 and 4.5.2.3).
(k) Dimensional sampling procedure approved by the procuring activity (see 4.6.2).
(l) If approval was requested and received for a different ballistic testing facility (see 4.7.4.1).
(m) Rejection and retest requirements, if different (see 4.8).
(n) If reduced testing is allowed (see 4.9).
(o) Packaging requirements (see 5.1).

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(p) Striking velocities for those thicknesses not covered (see MIL-STD-3057 and TOP 2-2-711).

6.3 Weldability. The U.S. Army DEVCOM, Ground Vehicle Systems Center, DEVCOM ARL, and the Aberdeen Test Center have conducted testing and determined that class II alloy can be welded and demonstrates good shock resistance when prepared and tested in accordance with MIL-STD-3057 and TOP 2-2-711. If these alloys in thicknesses required for manufacturing are not currently listed in TOP 2-2-711, the fabricator shall follow the procedures of TOP 2-2-711 to generate the required striking velocities before the Proofing Projectile Impact Testing (PPIT) is performed. The fabricator performing welding on these alloys for armor applications is responsible for developing and qualifying welding procedures in accordance with MIL-STD-3057 and TOP 2-2-711. Tests have demonstrated that material covered by this specification is weldable to itself and other weldable 6000 series alloys using a 4000 series filler and is weldable to 2000 series alloys using a 4000 series filler and to 5000 and 7000 series alloys using a 5000 series filler. It is not weldable to 2000 series alloys by conventional welding techniques. Any deviations from these conditions should be demonstrated prior to acceptance by the procuring activity.

6.4 Weld wire. Butt joints were successfully welded with AA4943 weld wire for class II (AA6055) material when tested with the required proofing projectile per MIL-STD-3057 (see 1.2).

6.5 Density. The density of AA6061 is 0.0975 lb/in^3 (2.70 g/cm^3). The density of AA6055 is 0.098 lb/in^3 (2.72 g/cm^3).

6.6 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, and pounds per square inch may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
Inch	0.0254	=	meter (m)
Foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.7 Subject term (key word) listing.

Ballistic testing
Caliber .30 AP M2
Caliber .50 AP M2
Caliber .50 FSP
M1114 HMMWV
Military vehicles
Stress corrosion
20-mm FSP

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

ARMOR PLATE, ALUMINUM ALLOY 6055 WELDABLE & ALLOY 6061, UNWELDABLE APPLIQUE

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of aluminum alloy armor plate, AA6061 and AA6055 when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification, and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V50 Ballistic Test for Armor

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity at a distance of at least two projectile diameters from any previous impact or disturbed area resulting from an impact, or from any crack, or from any edge of the test specimen.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any fair impact which is not a complete penetration may be considered a partial penetration.

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A.3.5 Witness plate. A witness plate is normally a 0.014 inch thick sheet of 5052-H36 aluminum alloy or a 0.020 inch thick sheet of 2024-T3 aluminum alloy placed 6 inches (+ 1/2 inch) behind and parallel to the test plates or other ballistic sample.

A.3.6 Spalling. The detachment or delamination of a layer of material in the area surrounding the location of impact, which may occur on either the front or rear surfaces of the armor. Ejected spall may impact the witness plate and cause a CP even when the projectile has not exited the plate.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V50 ballistic limit shall be in accordance with the values shown in tables A-I through A-VI.

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V50 Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP[P] velocity being higher than the low CP[P] velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

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A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at his expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered part of the lot they represent and remain the property of the producer. The now rejected lot also remains the property of the producer. The failed plates shall be returned, upon request, as in A.5.3.1.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
0.415	1217	1301	0.560	1438	1599
0.420	1225	1312	0.565	1445	1608
0.425	1233	1323	0.570	1452	1617
0.430	1241	1334	0.575	1459	1626
0.435	1249	1346	0.580	1466	1635
0.440	1257	1357	0.585	1473	1645
0.445	1265	1368	0.590	1480	1654
0.450	1273	1378	0.595	1487	1663
0.455	1281	1389	0.600	1494	1672
0.460	1290	1400	0.605	1500	1680
0.465	1298	1410	0.610	1507	1689
0.470	1305	1421	0.615	1514	1698
0.475	1313	1431	0.620	1521	1707
0.480	1321	1442	0.625	1527	1716
0.485	1328	1452	0.630	1534	1724
0.490	1336	1462	0.635	1541	1733
0.495	1343	1472	0.640	1547	1741
0.500 1/	1351	1483	0.645	1554	1750
0.505	1358	1493	0.650	1560	1758
0.510	1366	1503	0.655	1567	1767
0.515	1373	1512	0.660	1573	1775
0.520	1381	1522	0.665	1580	1784
0.525	1388	1532	0.670	1586	1792
0.530	1395	1542	0.675	1593	1800
0.535	1402	1551	0.680	1599	1809
0.540	1410	1561	0.685	1605	1817
0.545	1417	1570	0.690	1612	1825
0.550	1424	1580	0.695	1618	1833
0.555	1431	1589	0.700	1624	1841

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-I. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 30° obliquity - Continued

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
0.705	1630	1849	0.770	1711	1951
0.710	1637	1857	0.775	1717	1959
0.715	1643	1865	0.780	1723	1966
0.720	1649	1873	0.785	1728	1974
0.725	1655	1881	0.790	1734	1982
0.730	1661	1889	0.795	1739	1989
0.735	1668	1897	0.785	1745	1996
0.740	1674	1905	0.790	1751	2004
0.745	1680	1913	0.795	1757	2011
0.749 [2/]	1685	1919	0.800	1763	2019
0.750	1686	1920	0.805	1768	2026
0.755	1692	1928	0.810	1774	2033
0.760	1699	1936	0.815	1780	2041
0.765	1705	1944	0.820	1786	2048

[2/] Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
0.665	1535	1575	0.775	1848	1942
0.670	1547	1590	0.780	1863	1960
0.675	1560	1606	0.785	1879	1979
0.680	1573	1621	0.790	1895	1998
0.685	1586	1636	0.795	1912	2017
0.690	1599	1652	0.800	1928	2036
0.695	1612	1668	0.815	1978	2055
0.700	1625	1684	0.820	1995	2075
0.705	1639	1700	0.825	2012	2135
0.710	1653	1716	0.830	2029	2155
0.715	1667	1732	0.835	2046	2176
0.720	1681	1749	0.840	2064	2197
0.725	1696	1766	0.845	2081	2218
0.730	1711	1782	0.850	2099	2239
0.735	1725	1799	0.855	2117	2260
0.740	1740	1817	0.860	2135	2282
0.745	1755	1834	0.865	2153	2304
0.750 1/	1770	1852	0.870	2171	2326
0.755	1785	1869	0.875	2190	2348
0.760	1801	1887	0.880	2209	2370
0.765	1816	1905	0.885	2227	2393
0.770	1832	1923	0.890	2246	2416

1/ Specification requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits - caliber .50 fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
0.895	2265	2439	1.005	2726	3006
0.900	2284	2462	1.010	2749	3034
0.905	2304	2486	1.015	2772	3063
0.910	2323	2509	1.020	2795	3093
0.915	2343	2533	1.025	2819	3122
0.920	2363	2557	1.030	2842	3152
0.925	2383	2582	1.035	2866	3182
0.930	2403	2606	1.040	2890	---
0.935	2423	2631	1.045	2914	---
0.940	2444	2656	1.050	2939	---
0.945	2464	2682	1.055	2963	---
0.950 [2/]	2485	2707	1.060	2988	---
0.955	2506	2733	1.065	3012	---
0.960	2527	2759	1.065	3037	---
0.965	2549	2786	1.070	3062	---
0.970	2570	2812	1.075	3087	---
0.975	2592	2839	1.080	3113	---
0.980	2614	2866	1.085	3139	---
0.985	2636	2894	1.090	3165	---
0.990	2658	2921	1.095	3191	---
0.995	2680	2949	1.100	3217	---
1.000 [3/]	2703	2977	1.105	---	---

[2/] Specification class II requirements end for this ordered thickness.
[3/] Specification class I requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20 mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
0.865	---	1131	1.010	1435	1362
0.870	---	1138	1.015	1443	1371
0.875	---	1146	1.020	1450	1379
0.880	---	1153	1.025	1458	1388
0.885	---	1160	1.030	1466	1397
0.890	---	1168	1.035	1474	1406
0.865	---	1131	1.040	1481	1415
0.895	1271	1175	1.045	1489	1424
0.900	1277	1183	1.050	1497	1433
0.905	1284	1191	1.055	1505	1443
0.910	1291	1198	1.060	1513	1452
0.915	1298	1206	1.065	1521	1461
0.920	1305	1214	1.070	1529	1470
0.925	1311	1221	1.075	1537	1480
0.930	1318	1229	1.080	1546	1489
0.935	1325	1237	1.085	1554	1499
0.940	1332	1245	1.090	1562	1509
0.945	1339	1253	1.095	1570	1518
0.950	1346	1261	1.100	1579	1528
0.951 [1/]	1347	1263	1.105	1587	1538
0.955	1353	1269	1.110	1596	1548
0.960	1361	1277	1.115	1604	1558
0.965	1368	1286	1.120	1613	1568
0.970	1375	1294	1.125	1621	1578
0.975	1382	1302	1.130	1630	1588
0.980	1389	1310	1.135	1638	1598
0.985	1397	1319	1.140	1647	1608
0.990	1404	1327	1.145	1656	1619
0.995	1412	1336	1.150	1664	1629
1.000	1419	1344	1.155	1673	1640
1.001 [2/]	1420	1346	1.160	1682	1650
1.005	1427	1353	1.165	1691	1661

[1/] Specification class II requirements begin for this ordered thickness.
[2/] Specification class I requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20 mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
1.170	1700	1671	1.335	2022	2064
1.175	1709	1682	1.340	2033	2078
1.180	1718	1693	1.345	2043	2091
1.185	1727	1704	1.350	2054	2104
1.190	1736	1715	1.355	2065	2118
1.195	1745	1726	1.360	2076	2131
1.200	1755	1737	1.365	2087	2145
1.205	1764	1748	1.370	2098	2159
1.210	1773	1759	1.375	2109	2173
1.215	1783	1770	1.380	2120	2187
1.220	1792	1782	1.385	2131	2201
1.225	1801	1793	1.390	2142	2215
1.230	1811	1805	1.395	2153	2229
1.235	1821	1816	1.400	2165	2243
1.240	1830	1828	1.405	2176	2258
1.245	1840	1840	1.410	2187	2272
1.250	1849	1852	1.415	2199	2287
1.255	1859	1863	1.420	2210	2302
1.260	1869	1875	1.425	2222	2316
1.265	1879	1887	1.430	2234	2331
1.270	1889	1900	1.435	2245	2346
1.275	1899	1912	1.440	2257	2361
1.280	1909	1924	1.445	2269	2376
1.285	1919	1936	1.450	2281	2392
1.290	1929	1949	1.455	2293	2407
1.295	1939	1961	1.460	2305	2422
1.300	1949	1974	1.465	2317	2438
1.305	1959	1987	1.470	2329	2454
1.310	1970	1999	1.475	2341	2469
1.315	1980	2012	1.480	2353	2485
1.320	1991	2025	1.485	2366	2501
1.325	2001	2038	1.490	2378	2517
1.330	2012	2051	1.495	2390	2533

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - 20 mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
1.500 [3/]	2403	2550	1.665	2853	---
1.505	2415	2566	1.670	2867	---
1.510	2428	2583	1.675	2882	---
1.515	2441	2599	1.680	2897	---
1.520	2453	2616	1.685	2912	---
1.525	2466	2633	1.690	2928	---
1.530	2479	2650	1.695	2943	---
1.535	2492	2667	1.700 [4/]	2958	---
1.540	2505	2684	1.705	2973	---
1.545	2518	2701	1.710	2989	---
1.550	2531	2718	1.715	3004	---
1.555	2544	2736	1.720	3019	---
1.560	2558	2753	1.725	3035	---
1.565	2571	2771	1.730	3051	---
1.570	2585	2789	1.735	3067	---
1.575	2598	2807	1.740	3083	---
1.580	2612	2825	1.745	3100	---
1.585	2625	2843	1.750	3116	---
1.590	2639	2861	1.755	3132	---
1.595	2653	2879	1.760	3149	---
1.600	2666	2898	1.765	3165	---
1.605	2680	2917	1.770	3182	---
1.610	2694	---	1.775	3199	---
1.615	2708	---	1.780	3215	---
1.620	2722	---	1.785	3232	---
1.625	2737	---	1.790	3249	---
1.630	2751	---	1.795	3266	---
1.635	2765	---	1.800	3283	---
1.640	2780	---	1.805	3301	---
1.645	2794	---	1.810	3318	---
1.650	2809	---	1.815	3335	---
1.660	2838	---	1.820	---	---

[3/] Specification class II requirements end for this ordered thickness.
[4/] Specification class I requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
0.865	---	1771	1.015	1836	1982
0.870	---	1779	1.020	1842	1989
0.875	---	1786	1.025	1848	1995
0.880	---	1793	1.030	1854	2002
0.885	---	1801	1.035	1860	2009
0.890	---	1808	1.040	1866	2015
0.895	1681	1815	1.045	1872	2022
0.900	1687	1823	1.050	1878	2028
0.905	1694	1830	1.055	1884	2035
0.910	1701	1837	1.060	1890	2041
0.915	1707	1844	1.065	1896	2048
0.920	1714	1851	1.070	1902	2054
0.925	1720	1858	1.075	1907	2061
0.930	1727	1866	1.080	1913	2067
0.935	1733	1873	1.085	1919	2073
0.940	1740	1880	1.090	1925	2080
0.945	1746	1887	1.095	1931	2086
0.950	1753	1894	1.100	1937	2092
0.951 [1/]	1754	1895	1.105	1942	2099
0.955	1759	1901	1.110	1948	2105
0.960	1766	1908	1.115	1954	2111
0.965	1772	1914	1.120	1960	2117
0.970	1778	1921	1.125	1965	2124
0.975	1785	1928	1.130	1971	2130
0.980	1792	1935	1.135	1977	2136
0.985	1798	1942	1.140	1983	2142
0.990	1804	1949	1.145	1988	2148
0.995	1810	1955	1.150	1994	2155
1.000	1816	1962	1.155	2000	2161
1.001 [2/]	1817	1963	1.160	2005	2167
1.005	1823	1969	1.165	2011	2173
1.010	1829	1976	1.170	2016	2179

[1/] Specification class II requirements begin for this ordered thickness.
[2/] Specification class I requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
1.175	2022	2185	1.340	2198	2376
1.180	2027	2191	1.345	2203	2382
1.185	2033	2197	1.350	2208	2387
1.190	2038	2203	1.355	2213	2393
1.195	2044	2209	1.360	2218	2398
1.200	2050	2215	1.365	2223	2404
1.205	2055	2221	1.370	2229	2409
1.210	2060	2227	1.375	2234	2415
1.215	2066	2233	1.380	2239	2420
1.220	2071	2239	1.385	2244	2425
1.225	2077	2245	1.390	2249	2431
1.230	2082	2250	1.395	2254	2436
1.235	2088	2256	1.400	2259	2442
1.240	2093	2262	1.405	2264	2447
1.245	2098	2268	1.410	2269	2453
1.250	2104	2274	1.415	2273	2458
1.255	2109	2280	1.420	2278	2463
1.260	2114	2285	1.425	2283	2469
1.265	2120	2291	1.430	2288	2474
1.270	2125	2297	1.435	2293	2479
1.275	2130	2303	1.440	2298	2485
1.280	2136	2308	1.445	2303	2490
1.285	2141	2314	1.450	2308	2495
1.290	2146	2320	1.455	2313	2500
1.295	2151	2325	1.460	2318	2506
1.300	2157	2331	1.465	2322	2511
1.305	2162	2337	1.470	2327	2516
1.310	2167	2342	1.475	2332	2522
1.315	2172	2348	1.480	2337	2527
1.320	2177	2354	1.485	2342	2532
1.325	2183	2359	1.490	2347	2537
1.330	2188	2365	1.495	2352	2542
1.335	2193	2370	1.500 3/	2356	2548

3/ Specification class II requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
1.505	2361	2553	1.670	2514	---
1.510	2366	2558	1.675	2519	---
1.515	2371	2563	1.680	2523	---
1.520	2376	2568	1.685	2528	---
1.525	2380	2633	1.690	2532	---
1.530	2385	2650	1.695	2537	---
1.535	2390	2667	1.700	2541	---
1.540	2394	2684	1.705	2545	---
1.545	2399	2701	1.710	2550	---
1.550	2404	2718	1.715	2554	---
1.555	2409	2736	1.720	2559	---
1.560	2413	2753	1.725	2563	---
1.565	2418	2771	1.730	2568	---
1.570	2423	2789	1.735	2572	---
1.575	2427	2807	1.740	2576	---
1.580	2432	2825	1.745	2581	---
1.585	2437	2843	1.750	2585	---
1.590	2441	2861	1.755	2590	---
1.595	2446	2879	1.760	2594	---
1.600	2450	2898	1.765	2598	---
1.605	2455	2917	1.770	2603	---
1.610	2460	---	1.775	2607	---
1.615	2464	---	1.780	2611	---
1.620	2469	---	1.785	2616	---
1.625	2473	---	1.790	2620	---
1.630	2478	---	1.795	2624	---
1.635	2482	---	1.800	2629	---
1.640	2487	---	1.805	2633	---
1.645	2492	---	1.810	2637	---
1.650	2496	---	1.815	2641	---
1.655	2501	---	1.820	2646	---
1.660	2505	---	1.825	2650	---
1.665	2510	---	1.830	2654	---

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
1.835	2658	---	1.980	2779	---
1.840	2663	---	1.985	2783	---
1.845	2667	---	1.990	2787	---
1.850	2671	---	1.995	2791	---
1.855	2675	---	2.000 4/	2796	---
1.860	2680	---	2.005	2800	---
1.865	2684	---	2.010	2804	---
1.870	2688	---	2.015	2808	---
1.875	2692	---	2.020	2812	---
1.880	2697	---	2.025	2816	---
1.885	2701	---	2.030	2820	---
1.890	2705	---	2.035	2824	---
1.895	2709	---	2.040	2828	---
1.900	2713	---	2.045	2832	---
1.905	2717	---	2.050	2836	---
1.910	2722	---	2.055	2840	---
1.915	2726	---	2.060	2844	---
1.920	2730	---	2.060	2843	---
1.925	2734	---	2.065	2847	---
1.930	2738	---	2.070	2851	---
1.935	2742	---	2.075	2855	---
1.940	2746	---	2.080	2859	---
1.945	2751	---	2.085	2863	---
1.950	2755	---	2.090	2867	---
1.955	2759	---	2.095	2871	---
1.960	2763	---	2.100	2875	---
1.965	2767	---	2.105	2879	---
1.970	2771	---	2.110	2882	---
1.975	2775	---	2.115	2886	---

4/ Specification class I requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
1.390	---	1762	1.550	---	1899
1.395	---	1766	1.555	---	1903
1.400	---	1771	1.560	---	1907
1.405	---	1775	1.565	---	1912
1.410	---	1780	1.570	---	1916
1.415	---	1784	1.575	---	1920
1.420	---	1789	1.580	---	1924
1.425	---	1793	1.585	---	1928
1.430	---	1797	1.590	---	1932
1.435	---	1802	1.595	---	1936
1.440	---	1806	1.600	---	1940
1.445	---	1810	1.605	---	1944
1.450	---	1815	1.610	---	1948
1.455	---	1819	1.615	---	1952
1.460	---	1823	1.620	---	1956
1.465	---	1828	1.625	---	1960
1.470	---	1832	1.630	---	1964
1.475	---	1836	1.635	---	1968
1.480	---	1840	1.640	---	1972
1.485	---	1845	1.645	---	1976
1.490	---	1849	1.650	---	1980
1.495	---	1853	1.655	---	1984
1.500	---	1857	1.660	---	1988
1.501 1/	---	1858	1.665	---	1992
1.505	---	1862	1.670	---	1996
1.510	---	1866	1.675	---	2000
1.515	---	1870	1.680	---	2004
1.520	---	1874	1.685	---	2008
1.525	---	1878	1.690	---	2012
1.530	---	1883	1.695	---	2016
1.535	---	1887	1.700	---	2019
1.540	---	1891	1.705	---	2023
1.545	---	1895	1.710	---	2027

1/ Specification class II requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
1.715	---	2031	1.880	---	2155
1.720	---	2035	1.885	1995	2158
1.725	---	2039	1.890	1999	2162
1.730	---	2043	1.895	2002	2166
1.735	---	2046	1.900	2006	2169
1.740	---	2050	1.905	2009	2173
1.745	---	2054	1.910	2013	2177
1.750	---	2058	1.915	2016	2180
1.755	---	2062	1.920	2020	2184
1.760	---	2066	1.925	2023	2187
1.765	---	2069	1.930	2027	2191
1.770	---	2073	1.935	2030	2195
1.775	---	2077	1.940	2034	2198
1.780	---	2081	1.945	2037	2202
1.785	---	2084	1.950	2041	2205
1.790	---	2088	1.955	2045	2209
1.795	---	2092	1.960	2048	2212
1.800	---	2096	1.965	2051	2216
1.805	---	2099	1.970	2055	2219
1.810	---	2103	1.975	2058	2223
1.815	---	2107	1.980	2062	2226
1.820	---	2111	1.985	2065	2230
1.825	---	2114	1.990	2068	2234
1.830	---	2118	1.995	2072	2237
1.835	---	2122	2.000	2075	2241
1.840	---	2125	2.001 [2/]	2075	2241
1.845	---	2129	2.005	2079	2244
1.850	---	2133	2.010	2082	2248
1.855	---	2137	2.015	2085	2251
1.860	---	2140	2.020	2089	2255
1.865	---	2144	2.025	2092	2258
1.870	---	2148	2.030	2095	2261
1.875	---	2151	2.035	2099	2265

[2/] Specification class I requirements begin for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
2.040	2102	2268	2.205	2209	2380
2.045	2105	2272	2.210	2212	2383
2.050	2109	2275	2.215	2216	2386
2.055	2112	2279	2.220	2219	2390
2.060	2115	2282	2.225	2222	2393
2.065	2119	2286	2.230	2225	2396
2.070	2122	2289	2.235	2228	2400
2.075	2125	2292	2.240	2231	2403
2.080	2128	2296	2.245	2234	2406
2.085	2132	2299	2.250	2238	2409
2.090	2135	2303	2.255	2241	2413
2.095	2138	2306	2.260	2244	2416
2.100	2142	2310	2.265	2247	2419
2.105	2145	2313	2.270	2250	2422
2.110	2148	2316	2.275	2253	2426
2.115	2151	2320	2.280	2256	2429
2.120	2155	2323	2.285	2259	2432
2.125	2158	2326	2.290	2263	2435
2.130	2161	2330	2.295	2266	2439
2.135	2164	2333	2.300	2269	2442
2.140	2168	2337	2.305	2272	2445
2.145	2171	2340	2.310	2275	2448
2.150	2174	2343	2.315	2278	2451
2.155	2177	2347	2.320	2281	2455
2.160	2180	2350	2.325	2284	2458
2.165	2184	2353	2.330	2287	2461
2.170	2187	2357	2.335	2290	2464
2.175	2190	2360	2.340	2293	2467
2.180	2193	2363	2.345	2296	2470
2.185	2196	2367	2.350	2299	2474
2.190	2200	2370	2.355	2303	2477
2.195	2203	2373	2.360	2306	2480
2.200	2206	2377	2.365	2309	2483

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
2.370	2312	2486	2.540	2413	2591
2.375	2315	2489	2.545	2416	2594
2.380	2318	2493	2.550	2419	2597
2.385	2321	2496	2.555	2421	2601
2.390	2324	2499	2.560	2424	2604
2.395	2327	2502	2.565	2427	2607
2.400	2330	2505	2.570	2430	2610
2.405	2333	2508	2.575	2433	2613
2.410	2336	2511	2.580	2436	2616
2.415	2339	2515	2.585	2439	2619
2.420	2342	2518	2.590	2442	2622
2.425	2345	2521	2.595	2445	2625
2.430	2348	2524	2.600	2447	2628
2.435	2351	2527	2.605	2450	2631
2.440	2354	2530	2.610	2453	2634
2.445	2357	2533	2.615	2456	2636
2.450	2360	2536	2.620	2459	2639
2.455	2363	2539	2.625	2462	2642
2.460	2366	2543	2.630	2465	2645
2.465	2369	2546	2.635	2468	2648
2.470	2372	2549	2.640	2470	2651
2.475	2375	2552	2.645	2473	2654
2.480	2378	2555	2.650	2476	2657
2.485	2380	2558	2.655	2479	2660
2.490	2383	2561	2.660	2482	2663
2.495	2386	2564	2.665	2485	2666
2.500	2389	2567	2.670	2487	2669
2.505	2392	2570	2.675	2490	2672
2.510	2395	2573	2.680	2493	2675
2.515	2398	2576	2.685	2496	2678
2.520	2401	2579	2.690	2499	2681
2.525	2404	2582	2.695	2501	2684
2.530	2407	2585	2.700	2504	2687
2.535	2410	2588	2.705	2507	2690

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
2.710	2510	2692	2.845	2585	2770
2.715	2513	2695	2.850	2587	2773
2.720	2516	2698	2.855	2590	2776
2.725	2518	2701	2.860	2593	2779
2.730	2521	2704	2.865	2595	2781
2.735	2524	2707	2.870	2598	2784
2.740	2527	2710	2.875	2601	2787
2.745	2529	2713	2.880	2604	2790
2.750	2532	2716	2.885	2606	2793
2.755	2535	2719	2.890	2609	2795
2.760	2538	2721	2.895	2612	2798
2.765	2541	2724	2.900	2614	2801
2.770	2543	2727	2.905	2617	2804
2.775	2546	2730	2.910	2620	2807
2.780	2549	2733	2.915	2622	2809
2.785	2552	2736	2.920	2625	2812
2.790	2554	2739	2.925	2628	2815
2.795	2557	2742	2.930	2630	2818
2.800	2560	2744	2.935	2633	2821
2.805	2563	2747	2.940	2636	2823
2.810	2565	2750	2.945	2639	2826
2.815	2568	2753	2.950	2641	2829
2.820	2571	2756	2.955	2644	2832
2.825	2574	2759	2.960	2647	2835
2.830	2576	2762	2.965	2649	2837
2.835	2579	2764	2.970	2652	2840
2.840	2582	2767	2.975	2654	2843

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TABLE A-V. Minimum required ballistic limits - caliber .50 AP M2 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps		Thickness, inches	Required BL(P), fps	
	Class I	Class II		Class I	Class II
2.980	2657	2846	3.055	2693	2887
2.985	2660	2848	3.060	2695	2889
2.990	2662	2851	3.065	2697	2892
2.995	2665	2854	3.070	2699	2895
3.000 3/	2668	2857	3.075	2702	2897
3.005	2670	2859	3.080	2704	2900
3.010	2672	2862	3.085	2706	2903
3.015	2675	2865	3.090	2708	2906
3.020	2677	2868	3.095	2711	2908
3.025	2679	2870	3.100	2713	2911
3.030	2681	2873	3.105	2715	2914
3.035	2684	2876	3.110	2717	2916
3.040	2686	2878	3.115	2720	2919
3.045	2688	2881	3.120	2722	2922
3.050	2690	2884	3.125	2724	2924

3/ Specification requirements end for this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR

Preparing activity:
ARMY – MR

(Project 9535-2023-003)

Review activities:
Army - AR, AT, AV, TE
DLA - CQ

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